

24/09/25

1251110

Use matplotlib module for plotting in python.

AIM: To analyze and visualize the performance of student in different subject using matplotlib by plotting line, bar and pie charts.

ALGORITHM:

1. Import libraries

Import matplotlib.pyplot as plt and numpy for numerical operations (if needed)

2. Input Data

Store marks of 5 students in 3 subjects (Math, Science, English)

Using lists or arrays

3. Plot line chart

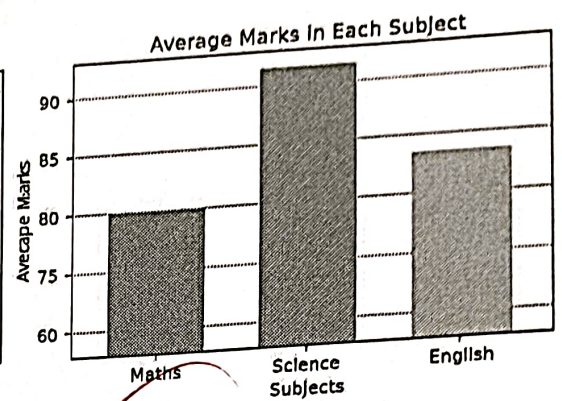
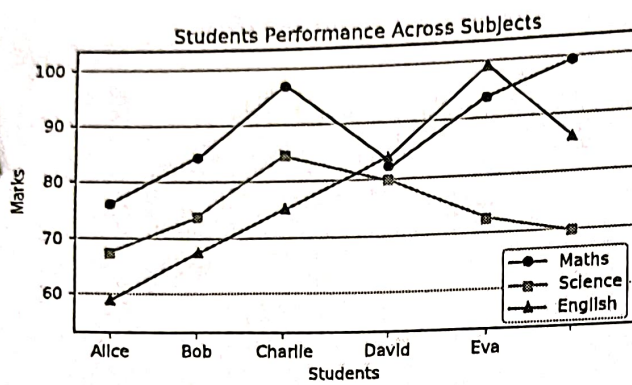
- X-axis $\rightarrow$ student
- Y-axis $\rightarrow$ marks in each subject
- Use `plt.plot()` for each subject with different colors and markers.
- Add title, labels, legend, and grid.

4. Plot Bar chart

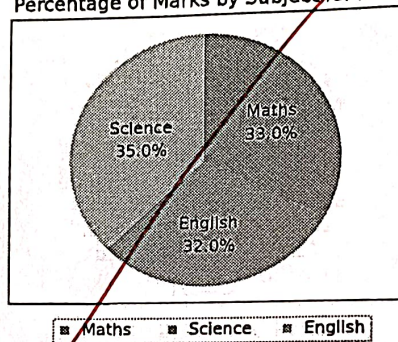
- Calculate average marks of each subject.
- Use `plt.bar()` to plot subject-wise averages.

5. Plot pie chart

- Select one student's marks.
- Use `plt.pie()` to show percentage contribution of marks in each subjects.



Percentage of Marks by Subject for Alice



prog 8910:-

```
import matplotlib.pyplot as plt
```

Data

```
Students = ['S1', 'S2', 'S3', 'S4', 'S5']
```

```
Maths = [85, 78, 92, 70, 88]
```

```
Science = [80, 75, 90, 68, 82]
```

```
English = [78, 82, 86, 74, 80]
```

1. Line chart

```
plt.figure(figsize=(10, 6))
```

```
plt.plot(Students, Maths, marker='o', label='Maths')
```

```
plt.plot(Students, Science, marker='s', label='Science')
```

```
plt.plot(Students, English, marker='^', label='English')
```

```
plt.title("Students' performance in different subjects")
```

```
plt.xlabel("Students")
```

```
plt.ylabel("Marks")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

2. Bar chart (Average marks per subject)

```
Subjects = ['Maths', 'Science', 'English']
```

```
avg_marks = [  
    sum(Maths) / len(Maths),  
    sum(Science) / len(Science),  
    sum(English) / len(English)
```

```
]
```


Sample output (visual description)

1. Line chart

- Three coloured lines showing marks of each subject for students S1-S5.
- Each subject line has different markers (circle, square + triangle).

2. Bar chart

- Three bars comparing average marks of Maths, science, and English

3. Pie chart

- Circular chart showing percentage contribution of each subject's marks for student S3.


```

plt.bar(subject: avg - marks_color = ['blue', 'green', 'orange'])
plt.title("Average marks in Each subject")
plt.xlabel("Subjects")
plt.ylabel("Average marks")
plt.grid(axis='y')
plt.show()

```

#3. Pie Chart for One Selected Student, say s3

```

student_index = 2 # s3
marks_s3 = [marks[student_index],
             science[student_index],
             english[student_index]]
plt.pie(marks_s3, labels = subjects, autopct = '%1.1f%%',
        ; startangle = 90)
plt.title("Percentage contribution of marks (student s3)")
plt.show()

```

VEL TECH - CSE	
EX NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

RESULT:

The program successfully visualized the student's performance using matplotlib.

15/10/25